AI-Powered Indoor Surveillance Robot
Senior Project Report - IT483 Systems Development Project I

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1. INTRODUCTION
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This project focuses on the development of an autonomous indoor surveillance robot using a Raspberry Pi 4 and computer vision technologies. The robot is designed to patrol the main hall of a university building, monitor the surrounding environment, detect human presence, and provide security alerts during restricted hours.

The system integrates embedded systems, computer vision, machine learning, and AI-assisted decision making. The robot processes visual data using a Raspberry Pi camera and OpenCV to understand its environment. It is capable of detecting human faces, recognizing key locations within the hall, and identifying exit areas to prevent leaving the designated patrol zone.

University buildings, particularly main halls, require continuous monitoring during restricted hours (after 10:00 PM) to ensure security and safety. Traditional security methods such as hiring guards for 24/7 surveillance are costly and prone to human error. This project proposes an intelligent, cost-effective alternative by deploying a mobile surveillance robot equipped with AI capabilities to autonomously patrol, detect, and alert security personnel of any unauthorized human presence.

This project is important for the university stakeholders because it addresses the need for enhanced campus security during off-hours while reducing the cost and dependency on human security personnel.

Document Structure:

- Chapter 1 introduces the project, its objectives, and the motivation behind it.
- Chapter 2 presents related works and existing solutions in the domain of surveillance robotics.
- Chapter 3 covers the preliminary investigation including company profile, problem statement, project scope, expected benefits and capabilities, development environment, feasibility study, and project planning.
- Chapter 4 presents the analysis and design of the system including risk analysis, business analysis, user requirements, and infrastructure analysis.

1.1 Project's Objectives

1.1.1 General Objective

To design, develop, and deploy an AI-powered autonomous indoor surveillance robot using Raspberry Pi 4 and computer vision technologies that can patrol a university main hall, detect human presence, and provide security alerts during restricted hours.

1.1.2 Specific Objectives

1. To design and build a mobile surveillance robot platform using Raspberry Pi 4 as the main controller with a two-wheel drive system and ultrasonic obstacle avoidance.

2. To implement real-time environment monitoring and face detection using a Raspberry Pi Camera Module and OpenCV computer vision library.

3. To develop and train a machine learning model for location/zone recognition inside the main hall, enabling the robot to identify its position based on visual features.

4. To implement a virtual boundary detection system using image classification that recognizes exit doors and outside areas, preventing the robot from leaving the designated patrol zone.

5. To develop an automated email alert system using SMTP protocol that captures an image and sends a notification when a human face is detected after 10:00 PM.

6. To integrate a LLaMA-based AI model for high-level decision support and environmental reasoning, combining local processing with AI-assisted interpretation.

7. To evaluate the system's performance in terms of detection accuracy, patrol coverage, response time, and reliability of alerts.

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2. RELATED WORKS

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Several existing solutions in the domain of indoor surveillance and security robotics were reviewed to understand current approaches and identify opportunities for improvement.

2.1 Knightscope K5 Autonomous Security Robot
Knightscope developed the K5 autonomous security robot, which is used in commercial environments such as malls and parking lots. The K5 uses a combination of sensors, cameras, and GPS for navigation and surveillance. It can detect anomalies and alert security personnel. However, it is designed for outdoor use, is commercially expensive, and requires a dedicated infrastructure for deployment.

2.2 Cobalt Robotics Indoor Security Robot
Cobalt Robotics offers an indoor security robot that patrols office buildings and corporate spaces. It combines autonomous navigation with human-in-the-loop monitoring, allowing remote operators to interact with people through the robot. While effective for corporate settings, it relies on expensive hardware and cloud-based services, making it less suitable for educational institutions with limited budgets.

2.3 OpenCV-Based Face Detection Systems
Numerous open-source projects utilize OpenCV for face detection and recognition. These systems have demonstrated the effectiveness of Haar cascades and deep learning-based detectors for real-time face detection on embedded platforms like Raspberry Pi. These projects typically focus on stationary camera setups rather than mobile robotic platforms.

2.4 Raspberry Pi-Based Surveillance Systems
Several academic and hobbyist projects have explored Raspberry Pi-based surveillance cameras with motion detection and email notification capabilities. These systems typically use the Pi Camera Module with OpenCV for motion and face detection. However, most are stationary and do not incorporate autonomous navigation or zone recognition.

2.5 LLaMA-Based AI Decision Systems
Recent developments in large language models, particularly the LLaMA family of models, have shown promise for edge-based AI reasoning. These models can be optimized for embedded deployment to provide contextual decision-making capabilities. Integration of such models with robotic systems is an emerging research area.

Advantages of the Proposed System:
Compared to existing solutions, this project offers several advantages:
- Lower cost by using Raspberry Pi 4 and open-source software.
- Combined mobile patrol and face detection capabilities.
- Zone recognition using custom-trained machine learning models.
- Virtual boundary enforcement without physical barriers.
- Automated email alerts for security events during restricted hours.
- Integration of a LLaMA-based AI model for intelligent decision support.
- Designed specifically for indoor university environments.

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3. PRELIMINARY INVESTIGATION

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3.1 Company Profile

Organization: Asia-Pacific International University (AIU)

Address: 195 Moo 3, Muak Lek, Saraburi 18180, Thailand
Line of Business: Higher Education Institution
Contact: [To be specified - Faculty of Information Technology office phone and email]

Stakeholders:
1. Security Department, AIU - Responsible for campus security
Contact: [To be specified]
2. Faculty of Information Technology - Academic supervision
Contact: [To be specified]
3. Building Management Office - Responsible for main hall operations
Contact: [To be specified]

3.2 Organizational Chart

[To be provided - An organizational chart showing the relevant departments of Asia-Pacific International University, with emphasis on the Security Department, Faculty of Information Technology, and Building Management Office and their relationships.]

3.3 Statement of the Mission/Goal of the Organization

Asia-Pacific International University aims to provide quality education in a safe and secure environment. The university is committed to leveraging technology to enhance campus operations, including security and facility management. The proposed surveillance robot aligns with the university's goal of adopting innovative solutions to improve campus safety and operational efficiency.

3.4 Project Request

This project was initiated as a Senior Project (IT483/IT484) at the Faculty of Information Technology, Asia-Pacific International University. The project was proposed by the student in response to the identified need for enhanced security monitoring of the university's main hall during restricted hours (after 10:00 PM). The project combines the student's interest in robotics, computer vision, and artificial intelligence to address a practical campus security challenge.

3.5 Description of the Problem

The main hall of the university building is a high-traffic area during the day but requires security monitoring during nighttime restricted hours (after 10:00 PM). Currently, security monitoring relies on periodic manual walkthroughs by security personnel, which is:

1. Inconsistent: Human patrols may miss certain areas or time periods.
2. Costly: Employing dedicated guards for 24/7 coverage requires significant personnel costs.
3. Limited in coverage: Guards cannot monitor all areas simultaneously.
4. Reactive rather than proactive: Human patrols react to incidents rather than continuously monitoring.
5. Subject to human fatigue: Guards working night shifts may experience fatigue, reducing alertness.

There is no automated system in place to continuously monitor the main hall, detect unauthorized human presence, and immediately alert security personnel. This creates a security gap that could lead to unauthorized access, theft, vandalism, or safety incidents going undetected.

The proposed AI-powered surveillance robot addresses this problem by providing continuous, autonomous patrol with real-time face detection and automatic alert capabilities.

3.6 Project Scopes and Constraints

IN-SCOPE:

Functions and Features:
- Autonomous indoor patrol within the university main hall using a two-wheel drive system.
- Real-time video capture and processing using Raspberry Pi Camera Module and OpenCV.
- Face detection using computer vision techniques (Haar cascades or deep learning-based models).
- Zone/location recognition inside the hall using a trained machine learning image classification model.
- Virtual boundary detection to prevent the robot from exiting the hall using image classification of exit doors and outside areas.
- Obstacle avoidance using HC-SR04 ultrasonic distance sensor.
- Automated email alerts via SMTP when faces are detected after 10:00 PM, including captured images and timestamps.
- AI-assisted decision making using a LLaMA-based model for high-level reasoning.

Data and Complexity:
- Input: Continuous video frames from Raspberry Pi Camera, ultrasonic sensor distance readings, time data.
- Output: Face detection results, zone classification, boundary detection results, captured alert images, email notifications, AI reasoning output.
- Training data: Labeled images from different zones within the main hall for the zone recognition model.

Users:
- Security Personnel: Receive email alerts and monitor robot activity.
- System Administrator: Configure robot parameters, email settings, and patrol schedules.

OUT-OF-SCOPE:
- The robot will not operate outdoors or in areas outside the main hall.
- The system will not perform facial recognition (identifying specific individuals); it only detects the presence of human faces.
- The robot will not have real-time video streaming to a remote monitoring dashboard (future improvement).
- The system will not integrate with existing university security infrastructure (CCTV, access control).
- The robot will not carry any defensive or deterrent mechanisms.

Quantitative Performance Requirements:
- The face detection system must achieve a detection rate of at least 85% at distances up to 3 meters in normal indoor lighting conditions.
- The zone recognition model must achieve a classification accuracy of at least 80% on test data.
- The ultrasonic sensor must detect obstacles at distances from 2 cm to 400 cm with accuracy of +/- 3 mm.
- The robot must complete a full patrol cycle of the main hall within 15 minutes.
- Email alerts must be sent within 30 seconds of face detection.
- The system must operate continuously for at least 2 hours on a single battery charge.

Constraints and Limitations:
- Technical: The project is constrained to using Raspberry Pi 4 as the main controller, which has limited computational power compared to dedicated GPUs. This affects the complexity of machine

learning models that can be deployed.
- Financial: The project operates on a limited student budget, restricting hardware choices to affordable components.
- Schedule: The project must adhere to the Senior Project Schedule, with IT483 deliverables due by week 15 and IT484 deliverables due by week 33.
- Environmental: The robot operates only in the indoor environment of the main hall with controlled lighting conditions.
- Power: Battery capacity limits continuous operation time.

Control Requirements:
- Access: The system requires administrator credentials to configure robot parameters and email settings.
- Roles: Two roles are defined: Security Personnel (receives alerts) and System Administrator (full configuration access).
- Audit: All face detection events, alerts sent, and patrol activities are logged with timestamps for audit purposes.

Mitigating Factors:
- OpenCV is a well-established, open-source library with extensive documentation for face detection on Raspberry Pi.
- Python is the primary programming language, which has strong support for all required libraries.
- The Raspberry Pi platform has a large community with extensive tutorials and support.
- Pre-trained face detection models (Haar cascades, DNN-based) are readily available.

3.7 Expected Business Benefits
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1. Enhanced Security Coverage: The robot provides continuous automated patrol and monitoring of the main hall during restricted hours, eliminating gaps in security coverage.

2. Cost Reduction: Reduces the need for additional security personnel for nighttime monitoring of the main hall, leading to long-term cost savings.

3. Immediate Alert Response: Automated email alerts with captured images enable security personnel to respond quickly to unauthorized presence, reducing response time compared to periodic manual checks.

4. Consistent Monitoring: Unlike human guards, the robot maintains consistent patrol patterns and detection performance without fatigue or distraction.

5. Evidence Collection: Captured images with timestamps provide documented evidence of security events for review and investigation.

6. Technology Demonstration: Demonstrates the practical application of AI and robotics in campus security, potentially leading to expanded deployment across the university.

7. Deterrence: The visible presence of a surveillance robot can serve as a deterrent to unauthorized access.

3.8 Expected System Capabilities
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1. Autonomous Navigation: The robot navigates through the main hall autonomously using its two-wheel drive system, following patrol patterns while avoiding obstacles detected by ultrasonic sensors.

2. Real-Time Face Detection: The system continuously processes video frames from the Pi Camera using OpenCV to detect human faces in real-time.

3. Zone Recognition: Using a trained machine learning image classification model, the robot identifies its current zone within the main hall (e.g., entrance area, central hall, corridor).

4. Virtual Boundary Enforcement: The robot recognizes exit doors and areas leading outside using image classification and automatically stops and changes direction to remain within the patrol zone.

5. Obstacle Avoidance: The HC-SR04 ultrasonic sensor detects obstacles in the robot's path, and the control system adjusts the robot's movement to avoid collisions.

6. Security Alert System: When a face is detected after 10:00 PM, the system captures a high-resolution image and sends an email alert via SMTP containing the image and detection timestamp.

7. AI-Assisted Decision Making: A LLaMA-based model processes environmental information and provides high-level reasoning to guide the robot's behavior in complex situations.

8. Activity Logging: All patrol activities, detections, and alerts are logged with timestamps for review and audit purposes.

3.9 Development Environment

Methodology:
The project follows an iterative development methodology. Each iteration focuses on integrating and testing specific components of the system, gradually building toward the complete integrated robot system.

Iteration 1: Hardware assembly and basic motor control.
Iteration 2: Camera integration and face detection implementation.
Iteration 3: Zone recognition model training and deployment.
Iteration 4: Virtual boundary detection implementation.
Iteration 5: Email alert system integration.
Iteration 6: LLaMA-based AI decision system integration.
Iteration 7: Full system integration and testing.

Modelling Languages:
- UML (Unified Modeling Language) for system design diagrams.
- Data Flow Diagrams (DFD) for process modeling.

Implementation Languages and Technologies:
- Python 3.x: Primary programming language for all software components.
- OpenCV (cv2): Computer vision library for image processing, face detection, and video capture.
- TensorFlow/Keras or scikit-learn: Machine learning frameworks for training zone recognition and boundary detection models.
- RPi.GPIO: Python library for Raspberry Pi GPIO pin control (motors, sensors).
- smtplib: Python library for SMTP email sending.
- LLaMA (via llama.cpp or similar): AI model for decision reasoning.
- Raspbian OS / Raspberry Pi OS: Operating system for Raspberry Pi 4.

Development Tools:
- Visual Studio Code with Remote SSH for code development.
- Git / GitHub (institutional repository) for version control.

- Jupyter Notebook for machine learning model development and testing.

3.10 Feasibility Study

3.10.1 Technical Feasibility

The project is technically feasible based on the following assessment:

Hardware Availability:
- Raspberry Pi 4 Model B is commercially available and has sufficient processing power (1.5 GHz quad-core ARM Cortex-A72, 4GB RAM) for running OpenCV and lightweight machine learning models.
- Raspberry Pi Camera Module V2 provides 8MP resolution suitable for face detection.
- HC-SR04 ultrasonic sensor is widely available and well-documented for use with Raspberry Pi.
- DC motors with motor driver modules (TB6612FNG or L298N) are standard components for robotics projects.

Software Availability:
- OpenCV provides well-established face detection algorithms (Haar cascades, DNN-based) with documented Raspberry Pi support.
- Python has extensive libraries for all required functionalities.
- Machine learning frameworks (TensorFlow Lite, scikit-learn) can run on Raspberry Pi.
- LLaMA models can be quantized and run on Raspberry Pi using llama.cpp.

Technical Skills:
- The student has foundational knowledge in Python programming, computer vision, and embedded systems from coursework.
- Extensive online resources and community support are available for all technologies used.

Conclusion: The project is technically feasible with available hardware, software, and skills.

3.10.2 Legal Feasibility

The project is legally feasible based on the following assessment:

- All software used is open-source or freely available for academic use (OpenCV: Apache 2.0 license, Python: PSF license, TensorFlow: Apache 2.0 license).
- The LLaMA model is available under Meta's community license for research and academic use.
- The robot operates within the university premises with authorization from the relevant departments.
- Face detection (not recognition) does not store personal biometric data of individuals, minimizing privacy concerns.
- The project complies with university policies regarding senior project requirements and institutional repository usage.
- Alert images are sent only to authorized security personnel.

Conclusion: The project is legally feasible with no identified legal barriers.

3.10.3 Operational Feasibility

The project is operationally feasible based on the following assessment:

- Security personnel require minimal training to understand and respond to email alerts.
- The robot operates autonomously, reducing the need for constant human intervention.
- The system administrator requires basic technical knowledge to configure the system parameters.
- The robot's patrol schedule can be configured to align with the university's restricted hours.

- Maintenance involves periodic battery charging and occasional hardware checks.
- The alert system uses standard email, which security personnel are already familiar with.

Conclusion: The project is operationally feasible and can be integrated into existing security workflows.

3.10.4 Schedule Feasibility

The project schedule aligns with the Senior Project Schedule:
- Weeks 1-4: Proposal submission and approval.
- Weeks 5-14: IT483 project execution (hardware assembly, face detection, zone recognition development).
- Week 15: IT483 deliverables submission.
- Weeks 16-19: IT483 defence preparation and presentation.
- Weeks 20-32: IT484 project execution (boundary detection, email alerts, LLaMA integration, full testing).
- Week 33: IT484 deliverables submission.
- Weeks 34-37: IT484 defence preparation and presentation.

Refer to Appendix - Software Cost Estimation for detailed schedule analysis.

Conclusion: The project schedule is feasible within the given academic timeline.

3.10.5 Financial Feasibility

Estimated Hardware Costs:
- Raspberry Pi 4 Model B (4GB): approximately 2,000 THB
- Raspberry Pi Camera Module V2: approximately 800 THB
- Robot chassis with 2 DC motors: approximately 500 THB
- Motor driver module (TB6612FNG): approximately 200 THB
- HC-SR04 ultrasonic sensor: approximately 50 THB
- Rechargeable battery pack: approximately 500 THB
- Voltage regulator and miscellaneous components: approximately 300 THB
- Total estimated hardware cost: approximately 4,350 THB

Software Costs:
- All software components are open-source: 0 THB

Operating Costs:
- Electricity for charging and development: minimal
- Internet for email alerts: university-provided

Total Estimated Project Cost: approximately 4,350 THB

Refer to Appendix - Software Cost Estimation for detailed financial analysis.

Conclusion: The project is financially feasible within a student budget.

3.11 Planning

A Gantt chart is to be provided showing the detailed breakdown of tasks aligned with the Senior Project Schedule (see Table 9 of the SDP Guidelines).

Phase 1 - IT483 (Weeks 1-19):
Week 1-4: Project proposal preparation and submission.

Week 5-6: Hardware procurement and robot chassis assembly.
Week 7-8: Motor control implementation and basic navigation.
Week 9-10: Camera integration and face detection using OpenCV.
Week 11-12: Data collection for zone recognition model.
Week 13-14: Zone recognition model training and initial testing.
Week 15: IT483 deliverables submission.
Week 16-19: IT483 defence preparation and presentation.

Phase 2 - IT484 (Weeks 20-37):
Week 20-22: Virtual boundary detection implementation.
Week 23-25: Email alert system development and testing.
Week 26-28: LLaMA-based AI decision system integration.
Week 29-31: Full system integration and comprehensive testing.
Week 32: Final testing and bug fixes.
Week 33: IT484 deliverables submission.
Week 34-37: IT484 defence preparation and presentation.

[A detailed Gantt chart should be included here.]

4. ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents the analysis and design of the AI-Powered Indoor Surveillance Robot system. The analysis covers risk assessment, business analysis including conduct of analysis and user requirements, and infrastructure analysis. The design phase addresses the physical design of the robot system and the software architecture.

The system is analyzed from multiple perspectives:
- Hardware perspective: Physical robot components and their interactions.
- Software perspective: Software modules, their responsibilities, and data flow.
- AI/ML perspective: Machine learning models, training data, and decision logic.
- Communication perspective: Email alert system and logging mechanisms.

4.2 Risk Analysis

The following risks have been identified and assessed for the project:

Risk 1: Insufficient Raspberry Pi Processing Power
- Probability: Medium
- Impact: High
- Mitigation: Use optimized algorithms and lightweight models. Use TensorFlow Lite for model inference. Reduce frame resolution if needed for real-time processing. Pre-process images before running detection.

Risk 2: Poor Face Detection Accuracy in Low Light
- Probability: Medium
- Impact: High
- Mitigation: Use DNN-based face detection which performs better in varying lighting. Consider adding LED illumination to the robot. Adjust camera exposure settings. Test extensively in target lighting

conditions.

Risk 3: Insufficient Battery Life for Patrol
- Probability: Medium
- Impact: Medium
- Mitigation: Optimize power consumption by reducing CPU usage during non-critical periods. Use higher-capacity battery. Implement power-saving patrol modes. Designate a charging station location.

Risk 4: Zone Recognition Model Inaccuracy
- Probability: Medium
- Impact: Medium
- Mitigation: Collect diverse training images under different lighting conditions. Use data augmentation techniques. Employ transfer learning from pre-trained models. Iteratively improve with additional training data.

Risk 5: Hardware Component Failure
- Probability: Low
- Impact: High
- Mitigation: Keep spare components available. Test hardware regularly. Implement software fault detection to gracefully handle sensor failures.

Risk 6: Network Connectivity Issues for Email Alerts
- Probability: Low
- Impact: Medium
- Mitigation: Implement email queue with retry mechanism. Store alert images locally as backup. Verify network connection before each alert attempt.

Risk 7: Schedule Delays
- Probability: Medium
- Impact: High
- Mitigation: Follow iterative development to deliver working increments early. Prioritize core features (motor control, face detection, alerts) over advanced features (LLaMA integration). Maintain regular progress reports with the adviser.

4.3 Business Analysis and Design

4.3.1 Conduct of Analysis

The analysis was conducted using the following methods:

1. Observation: The main hall of the university building was observed during different times of day and night to understand traffic patterns, lighting conditions, physical layout, and potential patrol routes. Key observations include:
- The hall has identifiable zones (entrance area, central hall, corridors) with distinct visual features.
- Lighting conditions vary between daytime (natural + artificial) and nighttime (artificial only).
- Exit doors and outside areas are visually distinguishable from interior areas.
- Obstacles include furniture, columns, and temporary objects.

2. Interview with Security Personnel: Security staff were consulted to understand current security procedures, pain points, and requirements for an automated surveillance system. Key findings:
- Current patrols are conducted every 2 hours during night shifts.
- The main hall is identified as a priority area for monitoring.
- Email is the preferred method for receiving alerts.
- Detection of any human presence after 10:00 PM is the primary concern.

3. Literature Review: Academic papers and existing projects on surveillance robotics, face detection on embedded systems, and indoor navigation were reviewed to inform system design decisions.

4. Prototyping: Initial prototypes were developed to test face detection performance on Raspberry Pi and motor control feasibility.

4.3.2 User Requirements

OUTPUT REQUIREMENTS:
- Type: Email alert notifications with captured images.
- Layout: Email contains subject line with "Security Alert", body with date, time, and zone location, and attached JPEG image of detected face.
- Frequency: Triggered immediately upon face detection after 10:00 PM. Maximum one alert per detection event.
- Log output: Text-based log files recording all patrol activities, detections, and system events with timestamps.

INPUT REQUIREMENTS:
- Data Capture: Continuous video frames from Raspberry Pi Camera Module (640x480 resolution at minimum 15 FPS).
- Distance readings from HC-SR04 ultrasonic sensor (triggered every 100ms during movement).
- System clock time for determining restricted hours.
- Input Devices: Raspberry Pi Camera Module V2 (visual data), HC-SR04 ultrasonic sensor (distance data).

PROCESS REQUIREMENTS:
- Face detection: Process each video frame through OpenCV face detection pipeline to identify human faces.
- Zone classification: Periodically classify current location using the trained ML model.
- Boundary detection: Continuously check for exit areas using image classification.
- Obstacle avoidance: Process ultrasonic sensor readings and adjust motor commands.
- Alert decision: Compare current time with restricted hours threshold (10:00 PM) and trigger alert if face detected.
- AI reasoning: Feed environmental data to LLaMA model for high-level behavior decisions.

PERFORMANCE REQUIREMENTS:
- Time: Face detection processing must complete within 200ms per frame. Email alerts sent within 30 seconds of detection.
- Speed: Robot patrol speed of approximately 0.3 m/s for safe indoor navigation.
- Volume: System must handle continuous video processing for at least 2 hours per patrol session.
- Schedule: System operates during restricted hours (10:00 PM to 6:00 AM) or as configured.

CONTROL REQUIREMENTS:
- Access: Administrator access required for system configuration (email settings, patrol parameters, restricted hours).
- Users: Security Personnel (alert recipients) and System Administrator (configuration access).
- Roles: Security Personnel role has read-only access to alerts and logs. Administrator role has full system configuration access.
- Audit: All detections, alerts, patrol routes, and system events are logged with timestamps in persistent log files.

USER TRAINING REQUIREMENTS:
- Security Personnel: Brief training (30 minutes) on understanding email alerts, interpreting captured images, and escalation procedures.
- System Administrator: Technical training (2 hours) on system configuration, startup procedures, maintenance tasks, and troubleshooting common issues.

CONSTRAINTS:
- Time: Project must be completed within the academic schedule (37 weeks total).
- Funds: Limited to student budget (approximately 4,350 THB for hardware).
- Skills: Single developer with foundational programming and electronics skills.
- Technology: Constrained to Raspberry Pi 4 platform capabilities.
- External Factors: University WiFi availability for email alerts, indoor lighting conditions.

4.3.3 Infrastructure Analysis

HARDWARE - Available vs Required:

Available:
- Personal computer for development
- University WiFi network
- University power outlets

Required:
- Raspberry Pi 4 Model B (4GB RAM) - To be procured
- Raspberry Pi Camera Module V2 - To be procured
- Robot chassis with 2 DC motors and wheels - To be procured
- Motor driver module (TB6612FNG or L298N) - To be procured
- HC-SR04 ultrasonic distance sensor - To be procured
- Rechargeable battery pack (7.4V or 12V) - To be procured
- Voltage regulator (5V for Raspberry Pi) - To be procured
- Jumper wires, breadboard, mounting hardware - To be procured

SOFTWARE - Available vs Required:

Available:
- Python 3.x (pre-installed on Raspberry Pi OS)
- Raspberry Pi OS (free download)
- Visual Studio Code (free)
- Git (free)

Required:
- OpenCV (cv2) - Free, open-source, to be installed via pip
- TensorFlow Lite or scikit-learn - Free, open-source, to be installed via pip
- RPi.GPIO - Free, pre-installed on Raspberry Pi OS
- smtplib - Free, included in Python standard library
- llama.cpp or equivalent - Free, open-source, to be compiled/installed
- NumPy, Pillow - Free, open-source, to be installed via pip

NETWORKING - Available vs Required:

Available:
- University WiFi network with internet access

Required:
- WiFi connectivity for Raspberry Pi (built-in WiFi on Pi 4)
- Access to SMTP server for sending email alerts (Gmail SMTP or university email server)
- Stable internet connection during patrol for email delivery

STANDARDS - Available vs Required:

Available:
- University IT policies and guidelines
- Senior Project guidelines (SDP Guidelines v544)

Required:
- Python PEP 8 coding standards
- OpenCV best practices for embedded systems
- Machine learning model evaluation standards (accuracy, precision, recall)
- Email security standards (TLS encryption for SMTP)

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APPENDIX
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Appendix A - Software Cost Estimation
[To be completed with detailed financial cost, human cost, and time estimation.]

Appendix B - Documentation
[To be completed during IT484.]
B.1 Program Documentation (data dictionary, DFD, object models, screen layouts)
B.2 Operations Documentation (installation process, user roles, scheduling)
B.3 User Documentation (manuals, help, tutorials, FAQs)